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COURSE:	**Autonomous Systems Lab 1**
GROUP:	**4**
SCENARIO:	**Platoon Splitting (TASK 3)**

REQUIRED DATA / SIGNALS / EVENTS FOR PLATOON SPLITTING COMMUNICATION

The trucks must exchange the following information to safely split the original platoon into two stable sub-platoons.

PHASE	REQUIRED DATA/SIGNALS	REQUIRED COMMUNICATION EVENTS
1. Detect upcoming divergence	Truck ID, current position, speed, heading, route plan, divergence-point coordinates, distance and time to divergence, intended exit or route	Divergence Detected, Route Intent Broadcast, Divergence Acknowledged
2. Initiate manual override	Driver override command, override status, requesting truck ID, timestamp, communication mode, reason for override	Manual Override Requested, Override Accepted or Override Rejected, Manual Control Activated
3. Coordinate platoon separation	Current platoon order, intended sub-platoon membership, selected second leader, assigned route for each truck, planned split time and location	Split Request, Split Plan Broadcast, Sub-Platoon Assignment, Leader Nomination, Leader Acceptance, Split Plan Confirmed

PHASE	REQUIRED DATA/SIGNALS	REQUIRED COMMUNICATION EVENTS
4. Create safe gaps	Vehicle speed, acceleration, deceleration, longitudinal position, inter-vehicle distance, desired gap, braking capability, time headway, gap-control status	Gap Creation Start, Speed Adjustment Command, Gap Target Reached, Gap Creation Failed
5. Split control groups	New leader ID, follower IDs, sub-platoon ID, control authority, route assignment, predecessor vehicle ID, communication group/channel	Leadership Transfer, Control Authority Accepted, Leave Original Platoon, Join New Sub-Platoon, Control Group Activated
6. Reestablish V2V communication	Communication address, channel, signal strength, latency, packet-loss status, synchronization timestamp, heartbeat status	Connection Request, Connection Accepted, V2V Link Established, Heartbeat Received, Communication Timeout
7. Stabilize both sub-platoons	Actual and target gap, relative speed, acceleration, heading, lane position, controller state, stability indicator, fault status	Gap Synchronization Start, Stable Gap Reached, Sub-Platoon Stable, Stability Timeout, Emergency Fallback Activated

Essential continuous vehicle-state signals

These signals should be transmitted periodically during the complete operation:

- Vehicle identifier and role: TruckID, Leader, or Follower
- Position, lane, heading, and route
- Speed and acceleration
- Brake and throttle status
- Distance and relative speed to the preceding truck
- Desired inter-vehicle gap and actual gap
- Platoon ID and sub-platoon ID
- Current operating mode: automated, manual override, splitting, or stabilized
- V2V connection quality and heartbeat
- Fault, hazard, or emergency-braking status

Key command and acknowledgement signals

The interaction should use command acknowledgement pairs to ensure that every truck has received and accepted critical instructions:

1. Split Request → Split Request Acknowledged
2. Leader Nomination → Leader Role Accepted
3. Sub-Platoon Assignment → Assignment Confirmed
4. Gap Adjustment Command → Gap Target Reached
5. Leadership Transfer → Control Authority Accepted
6. V2V Reconnection Request → V2V Link Established
7. Stabilization Command → Sub-Platoon Stable

Resulting communication sequence

The lead truck broadcasts the detected divergence and the intended split plan. Each follower confirms its assigned route and sub-platoon, after which the selected follower accepts leadership of the second group. The trucks then exchange speed, acceleration, position, and gap data to create safe separation, transfer control authority, establish new V2V communication groups, and confirm that both resulting sub-platoons have reached stable spacing and synchronized control.